**EXP. NO: 22**

# SIMULATION AND ANALYSIS OF BPSK MODULATION USING CONSTELLATION DIAGRAMS IN MATLAB

## Objectives

1. To simulate Binary Phase Shift Keying (BPSK) modulation in MATLAB for a given binary data sequence.
2. To generate and analyze the BPSK constellation diagram and identify the two signal points corresponding to binary 0 and binary 1.
3. To study the effect of channel noise/AWGN on the BPSK constellation and evaluate signal quality using constellation parameters.
4. To compare the transmitted and detected symbols and determine the performance of the BPSK communication system.

## CODE

```
clc;
clear;
close all;

% Generate random bits
N = 100;
bits = randi([0 1], N, 1);

% BPSK modulation: 0 -> -1, 1 -> +1
bpsk = 2*bits - 1;

% Add AWGN noise manually (SNR = 10 dB)
SNR = 10;
noise_power = 10^(-SNR/10);
noise = sqrt(noise_power) * randn(N,1);
rx = bpsk + noise;

% Add small imaginary noise for constellation display
rx_complex = rx + 1i * sqrt(noise_power) * randn(N,1);

% Plot results
figure;

subplot(2,1,1);
stem(bits);
title('Input Binary Data');
```

```
xlabel('Bit Number');
ylabel('Bit');
grid on;

subplot(2,1,2);
scatter(real(rx_complex), imag(rx_complex), 'filled');
title('BPSK Constellation Diagram with AWGN');
xlabel('In-Phase');
ylabel('Quadrature');
grid on;
axis([-2 2 -2 2]);
```

# OUTPUT

